

Proforma Hypothesis Testing

TL;DR — What This Analysis Is About

1. Do promotions actually work?

When a car wash site suddenly spends a lot more than usual in a month (on discounts, marketing, or deals), we track what happens over the next year. The data shows that after a big spend, more customers sign up for memberships, and the site's revenue grows — and that growth sticks around for 12+ months, not just during the promo.

2. Are they winning new customers, or just stealing from competitors?

When one site runs a promo and gains members, where do those customers come from? We checked all other sites within 20km. The answer: neighboring sites lose retail customers at the same time. About 4 in 10 nearby sites see their revenue drop. This means the promo is mostly redistributing existing car wash customers, not creating new ones.

3. Is the spend worth it financially?

We looked at the actual dollars - what the site spent on the campaign, what revenue it earned, and what profit came out.

1. OPEX Spike (Campaigns)

A car wash site's operating expenses (OPEX) occasionally jump well above their normal level for a month or two or three. We treat these jumps as promotional events - the site is likely running a discounted membership offer or a marketing push. The event study tracks what happens to the site's own revenue, membership sign-ups, and wash counts in the 6 months before the spike through the 12 months after. The resulting chart shows whether promotions actually convert retail customers into members and whether that effect persists.

Event Definition

- **OPEX spike** = any month where a site's `total_expenses` exceeds $1.2\times$ its own 6-month trailing moving average
 - Threshold $1.2\times$ chosen to exclude normal month-to-month variance while capturing deliberate spend increases
 - **Sample:** 264 spike months across 97 of 162 sites in the data
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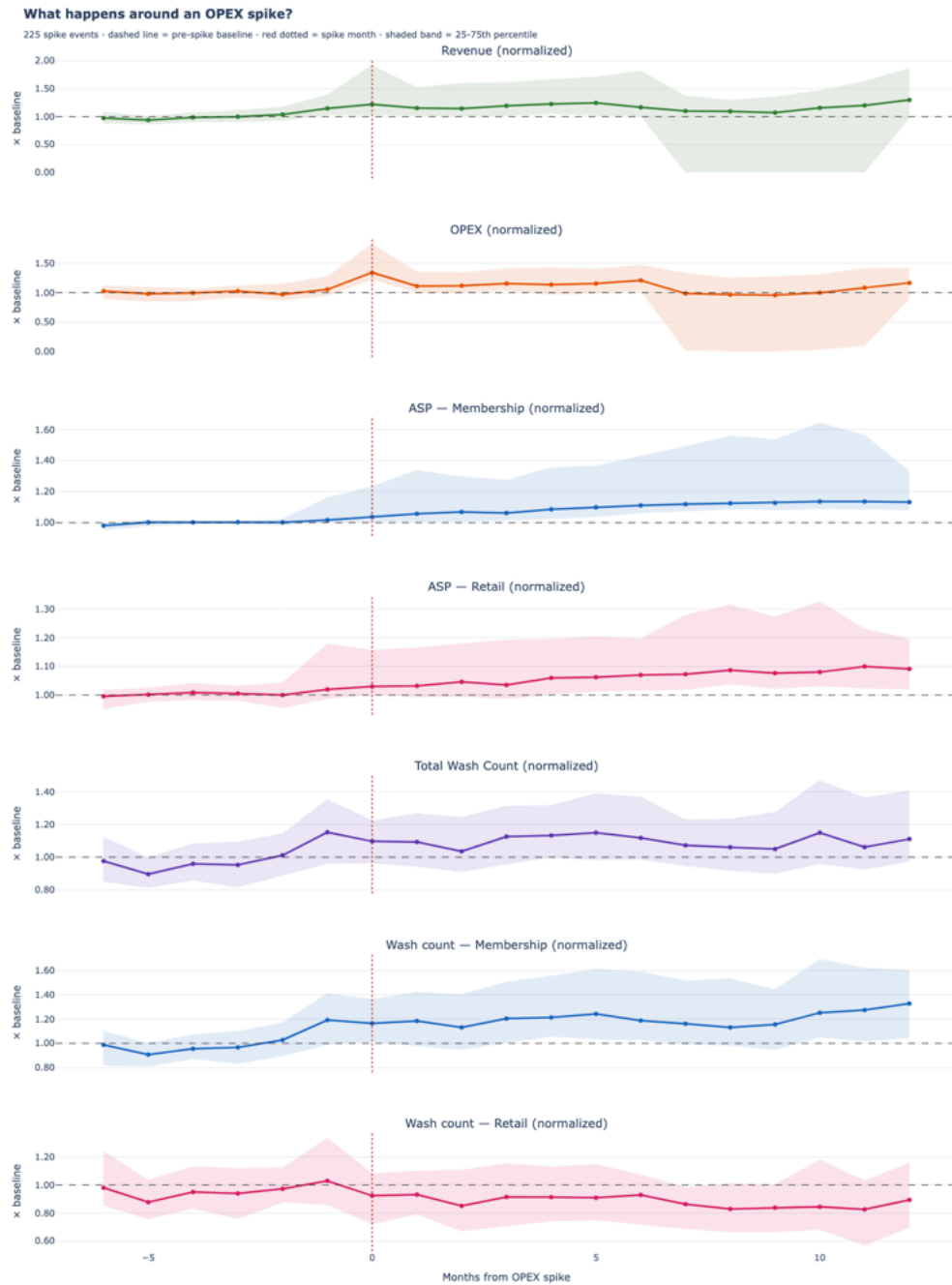
Methodology

- **Window:** $T = -6$ to $T = +12$ months relative to the spike month ($T = 0$)

- **Normalization:** each site's metrics divided by that site's own pre-spike median ($T = -6$ to $T = -1$). This makes sites of different sizes comparable — 1.0 always means "at baseline."
- **Metrics tracked:** OPEX, total revenue (`total_income`), membership ASP (`ASP_mem`), retail ASP (`ASP_ret`), membership wash count (`mem_wash_count`), retail wash count (`ret_wash_count`)
- **Aggregation:** at each month offset, take the **median** of normalized values across all spike events. Shaded band = 25th–75th percentile across events.
- Median of per-event % changes is used throughout (not % change of the medians), so each spike event gets equal weight regardless of site size.

How to Read the Chart

- **x-axis:** months from the spike ($T = 0$ is the spike month itself; negative = before; positive = after)
- **y-axis:** metric as a multiple of pre-spike average (1.0 = normal, 1.2 = 20% above normal)
- **Shaded band:** shows how wide the spread is — a narrow band means sites behave consistently; a wide band means outcomes vary a lot
- **What to look for:** `ASP_mem` typically drops at $T = 0$ (discount signal), `mem_wash_count` rises from $T = +1$ onwards (conversion signal), `ret_wash_count` falls post-spike (retail customers converted to members)



2. Impact of Focal OPEX spike on Neighbours

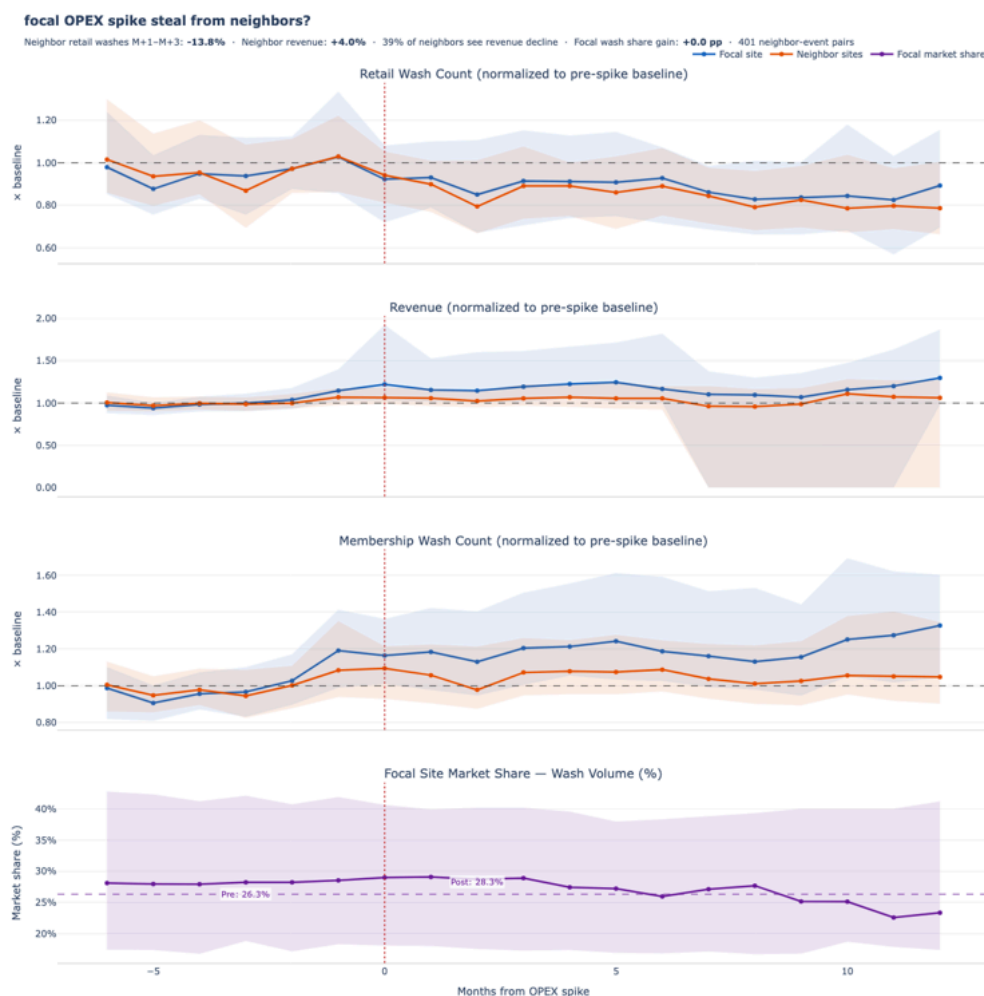
When a site runs a promotion and gains customers, two things could be happening: it could be pulling in entirely new customers who weren't washing cars before (market growth), or it could be taking customers away from nearby competitors (market redistribution). To distinguish between these, we track what happens to all other sites within 20km during the same period. If neighboring sites lose retail customers at the same time the focal site gains them, that is strong evidence of stealing — not growth.

Event Definition

- Same 264 spike events as Section 1
- **Neighbor** = any site within 20km straight-line distance of the focal site (Haversine formula applied to lat/lon coordinates)
- **Sample**: 448 unique focal-neighbor pairs across 106 focal sites with at least one neighbor; 388 unique (focal site, spike date, neighbor) triplets with sufficient data

Methodology

- For each spike event, collect the same $T = -6$ to $T = +12$ window of data for every neighboring site
- Each neighbor is normalized by **its own** pre-spike baseline ($T = -6$ to $T = -1$), not the focal site's baseline. This ensures a neighbor's size doesn't affect whether it looks like it gained or lost.
- **Post-window statistic**: median percentage change over $M+1$ to $M+3$ (the three months immediately after the spike), averaged within each (focal, spike, neighbor) triplet first, then across all triplets



3. OPEX · Revenue · Profit · Membership Purchases Around Campaigns

A single spike month and a three-month promotional campaign are very different things financially. We group consecutive spike months into campaigns, then look at the raw dollar values of OPEX, revenue,

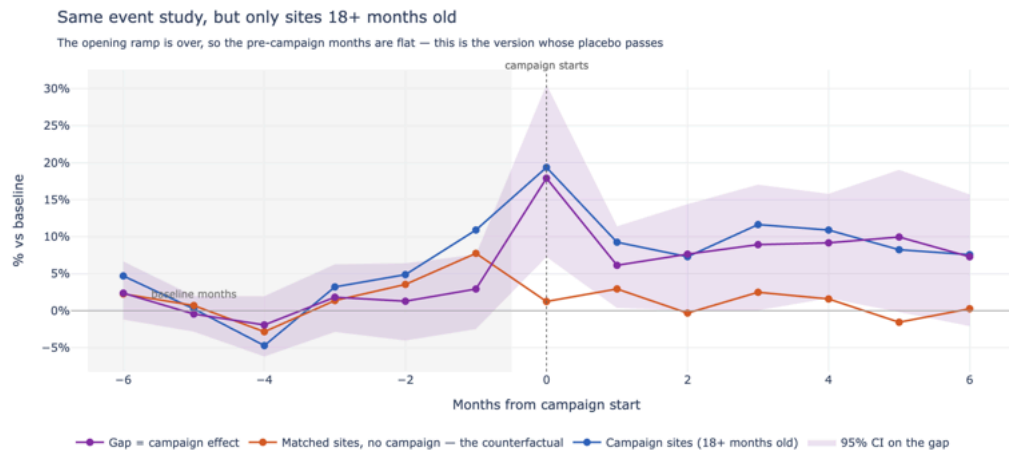
profit, and membership sign-ups across the campaign window. A separate statistical test answers the core question: when the campaign ends and OPEX drops, do profit and membership purchases reliably rise?

Event Definition

- **Campaign:** consecutive spike months for the same site with a gap of ≤ 1 month between them are merged into a single campaign
- 264 raw spike months → **184 campaigns** across 97 sites
- **Duration buckets:**
 - 1 month: 141 campaigns (76.6%)
 - 2 months: ~27 campaigns
 - 3+ months: ~16 campaigns
- **Anchor:** `campaign_start` (first spike month of the campaign)
- **Window:** $T = -3$ to $T = +6$ months relative to anchor; campaign months shaded yellow in all charts



3.2 Is this because of seasonality ? – Not strictly



The design compares growth at sites with an OPEX spike against sites without one, to reduce the effect of seasonality.

In the plot, the orange line (neighbour site) is the seasonal baseline: sites moving through the same calendar months as the campaign site. The result is a calendar-aligned plot centred on the OPEX spike of the focal site (blue line). The purple gap (blue – orange) is what remains once that shared movement is subtracted – so seasonality cancels out.

The **+7% lift** [+0.4%, +13.5%] on mature sites survives seasonality adjustment.

Summary

We checked whether the revenue jump after a promo is real or just good timing. It's real:

- Car wash seasonality is small ($\pm 10\%$)
- Removing the calendar effect entirely changes the answer by 5–6 points for sites with OPEX spike.
- Sites that ran no campaign, over the same months, stayed flat while campaign sites rose

4. Inception-Normalized Time Series (All Sites Aligned)

What it shows: How expense ratios evolve relative to each site's own operational lifecycle, regardless of when they opened.

Method:

- Each site's first `report_date` is treated as **Month 0** (month of inception)
- All subsequent months are re-indexed as Month 1, 2, 3, ... from that origin
- Sites that opened in April and sites that opened in October are both represented starting at Month 0, making them directly comparable
- The same median + P25/P75 aggregation is applied across sites at each lifecycle month

